TEXTBOOK EXERCISES

I. Choose the correct answer.

1. If Dobereiner is related with 'law of triads', then Newlands is related with
   a) Modern periodic law b) Hund's rule
   c) Law of octaves
   d) Pauli's Exclusion principle
2. Modern periodic law states that the physical and chemical properties of elements are the periodic functions of their _____
   a) atomic numbers b) atomic masses
   c) similarities d) anomalies
3. Elements in the modern periodic table are arranged in _____ groups and _____ periods.
   a) 7, 18 b) 18, 7 c) 17, 8 d) 8, 17

II. Fill in the blanks.

1. In Dobereiner's triads, the atomic weight of the middle element is the _____ of the atomic masses of 1st and 3rd elements.
2. Noble gases belong to _____ group of the periodic table.
3. The basis of the classifications proposed by Dobereiner, Newlands and Mendeleev was _____.
4. Example for liquid metal is _____.

III. Match the following.

Triads	Newlands
Alkali metal	Calcium
Law of octaves	Henry Moseley
Alkaline earth metal	Sodium
Modern Periodic Law	Dobereiner

IV. State whether true or false. If false, correct the statement.

1. Newlands' periodic table is based on atomic masses of elements and modern periodic table is based on atomic number of elements.
2. Metals can gain electrons.
3. Alloys bear the characteristics of both metals and nonmetals.
4. Lanthanides and actinides are kept at the bottom of the periodic table because they resemble each other but they do not resemble with any other group elements.
5. Group 17 elements are named as Halogens.

V. Assertion and reason type questions.

Statement: Elements in a group generally possess similar properties but elements along a period have different properties.

Reason: The difference in electronic configuration makes the element differ in their chemical properties along a period.

a) Statement is true and reason explains the statement.

b) Statement is false but the reason is correct.

VI. Answer the following.

1. State modern periodic law.
2. What are groups and periods in the modern periodic table?
3. What are the limitations of Mendeleev's periodic table?
4. State any five features of modern periodic table.

REFERENCE BOOKS

1. CONCISE Inorganic chemistry : 5th Edition by J.D. Lee
2. Inorganic Chemistry by P.L.Soni
3. The Periodic table: Its story and its significance: Eric R. Scerri

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